

INTRODUCTION

The transition to a weather-driven energy system presents significant forecasting challenges—also in the Austrian complex Alpine terrain, and especially with a national goal of achieving 100% electricity supply from renewable by 2030. This requires a substantial scale-up of wind and solar energy, on top of an already high share of hydropower. Accurate and actionable forecasts across multiple timescales are crucial to ensure grid stability, efficient market integration, and risk mitigation.

To address these challenges, the IEA Wind TCP Task 51 promotes international collaboration on renewable energy forecasting for the weather-driven energy system. The initiatives most relevant for Austria were for this reason introduced to a broad audience in an Austrian stakeholder workshop in Vienna in November 2024. The event brought together over 120 participants from more than 50 organizations spanning research, energy providers, system operators, and policymakers.

This poster presents key insights from the workshop, highlighting stakeholder needs, technological priorities, and implementation gaps in forecasting for Austria's weather-driven energy future.

KEY FINDINGS

FORECAST PRIORITIES

Stakeholders clearly emphasized that:

- **day-ahead forecasts** remain the most critical for operations (56%), aligning with market and grid stability needs.
- **nowcasting (42%)** and **intraday (35%)** are gaining relevance, especially for real-time trading and balancing.
- **weekly and sub-seasonal forecasts ("weeks++") were rated important by 31%**, pointing to a growing interest in **mid-range planning**—for storage scheduling, maintenance, and hedging against renewable variability.
- Lower ranking of **minute-scale forecasts (16%)** suggest current systems or operational value not yet clear.

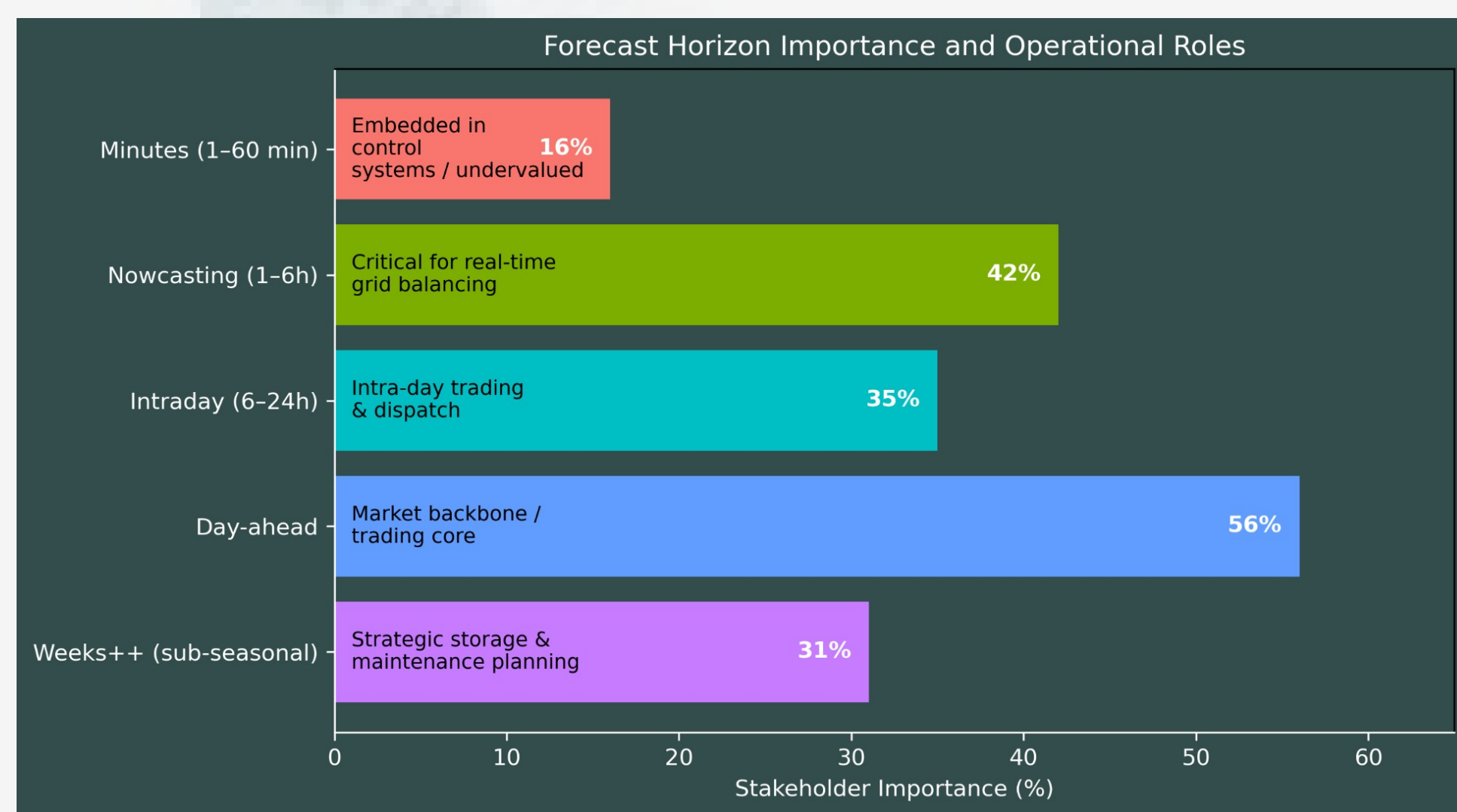


Figure 2. Forecast horizon relevance ranked by Austrian energy stakeholders. Day-ahead remains dominant, while short- and mid-range forecasts gain importance

WORKSHOP STRUCTURE

The first Austrian IEA Wind Task 51 workshop was held in Vienna on November 6, 2024:

- 120 participants from over 50 institutions.
- keynote presentations with interactive formats — Slido polling, World Café discussions, and Q&A rounds
- 4 thematic areas critical to energy transition in Alpine regions:
 1. Forecast horizons (nowcasting to seasonal)
 2. Extreme weather events
 3. Artificial intelligence & data science
 4. Open session for user-driven discussion

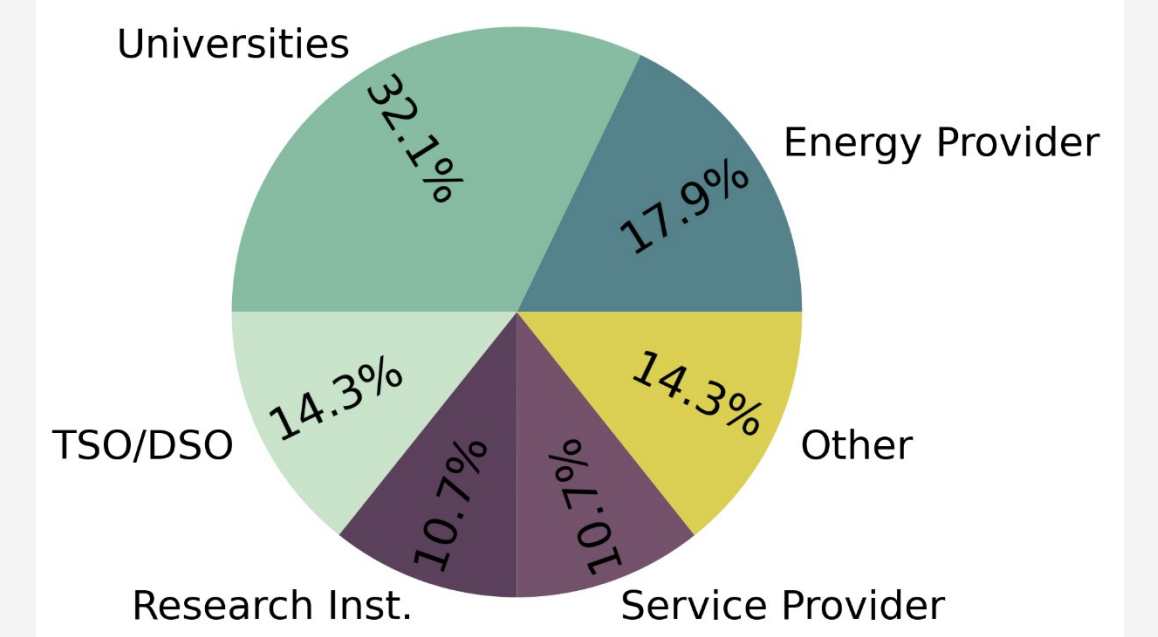


Figure 1. Participants per sector.

CLIMATE AND EXTREMES

- Extreme weather was rated as the **top concern** by 85% of participants, with particular focus on **storms, floods, heatwaves**, and compound risks such as **"Dunkelflaute"**.
- Awareness of **cascading effects** — where one event (e.g., storm) triggers another (e.g., power outage, market imbalance), amplifying systemic risks increased.
- Combined and consecutive events were perceived as particularly **difficult to forecast**.
- Climate change is expected to **increase the frequency and intensity** of such events

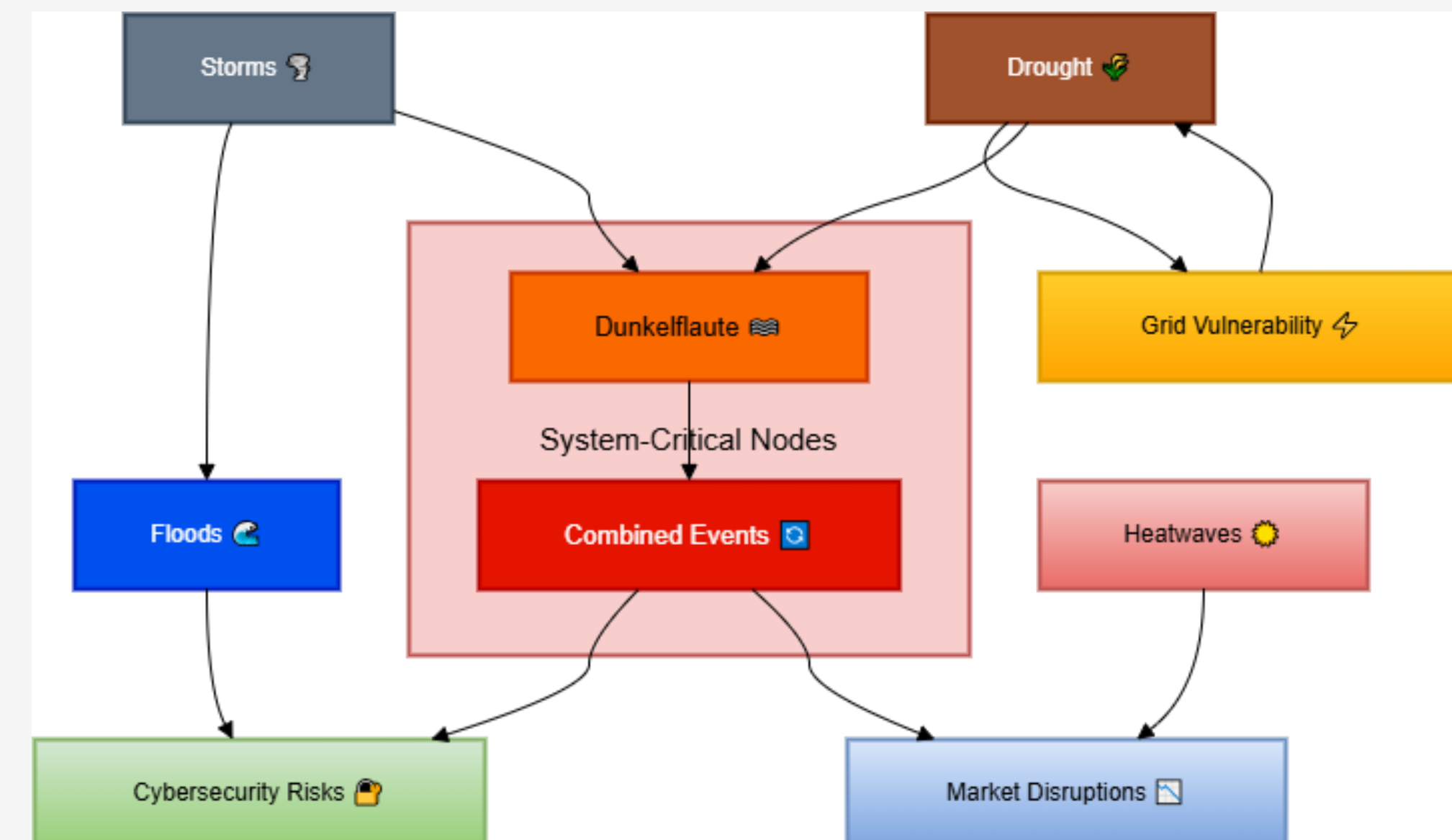


Figure 3. Stakeholder perception of climate and systemic risks. Dunkelflaute and compound events emerge as critical threats with cascading impacts.

AI & TRANSPARENCY

Artificial Intelligence (AI) is increasingly seen as a **game changer for renewable energy forecasting**, especially in complex Alpine terrain.

Stakeholders at the workshop rated the **importance of transparency in AI tools** as *very high* (4.35/5, SD 0.85).

This consensus highlights a demand for **interpretable, traceable, and validated** AI systems.

Participants specifically called for:

- **Standardized interfaces** for data and model exchange
- **Clear documentation** of AI decisions
- **Hybrid approaches** combining domain knowledge with machine learning
- **Continuous benchmarking** against conventional forecasting tools

Interestingly, cybersecurity **risks were frequently** linked to AI-driven automation and extreme weather threats, reflecting rising concerns over system vulnerability, reflecting a **shift of interest** from "black box" tools **toward AI governance** relevant in the **critical infrastructure** context of the energy system.

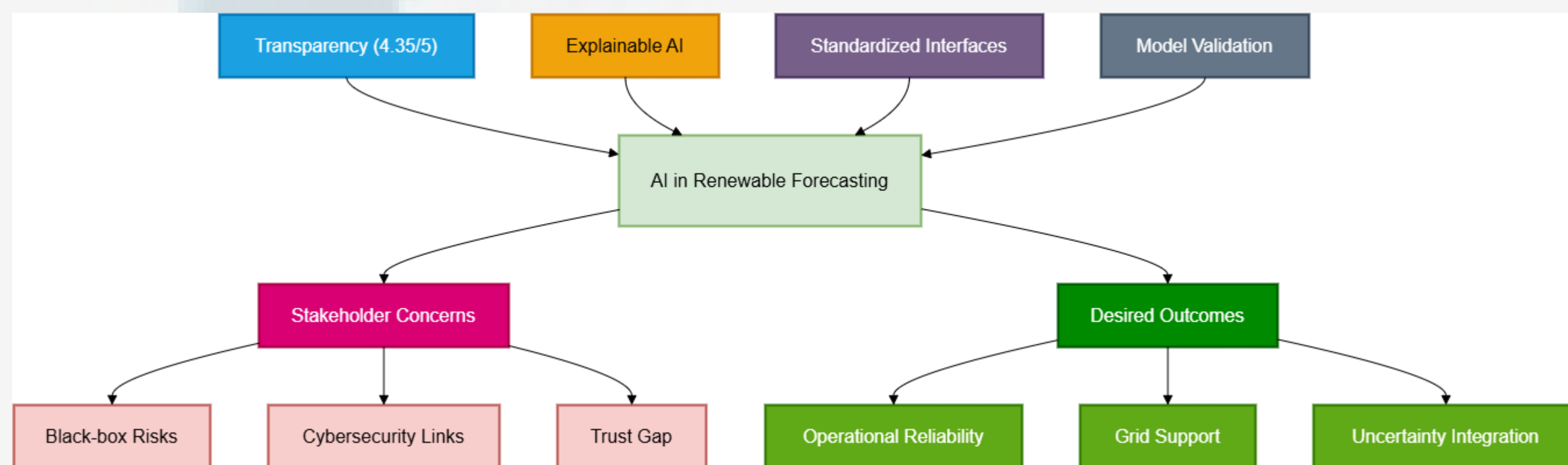


Figure 4. Stakeholder expectations for AI in forecasting: Transparency, explainability, and benchmarking are critical to enable operational trust.

UNCERTAINTY COMMUNICATION

Workshop showed a **gap between recognition and implementation of uncertainty & risk evaluation** remains.

Stakeholders rated **uncertainty communication 4.4/5** in importance, with **55% calling it "very important"** for new AI-based forecasts. In real-world practice, many users still rely on **gut feeling or past experience** though, especially in the absence of intuitive visualizations or standards.

Key barriers identified:

- Lack of **standardized uncertainty definitions & formats**
- Lack of intuitive **visual tools** tailored to decision-makers
- Limited examples showing **added value** in operations

This highlights a need for **sector-specific communication pathways** and better integration of uncertainty into daily workflows — particularly for extreme events and novel applications such as AI products.

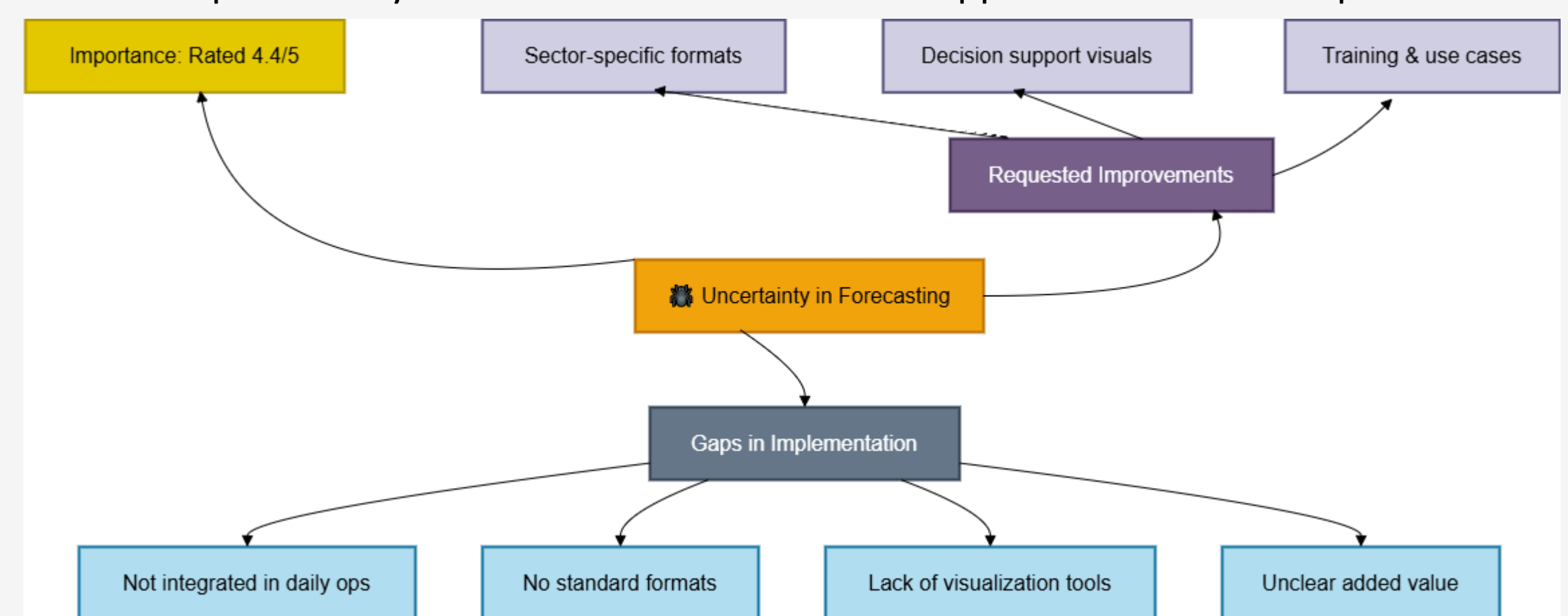


Figure 5. Uncertainty is seen as central to modern forecasting—but poorly implemented. Stakeholders call for visualization tools, standards, and targeted training.

STAKEHOLDER PRIORITIES & BARRIERS

Stakeholders expressed clear forecasting priorities — but also exposed implementation gaps. This layered system connects needs, blockers, and proposed solutions, guiding the path toward actionable improvements in Austria's energy forecasting landscape.

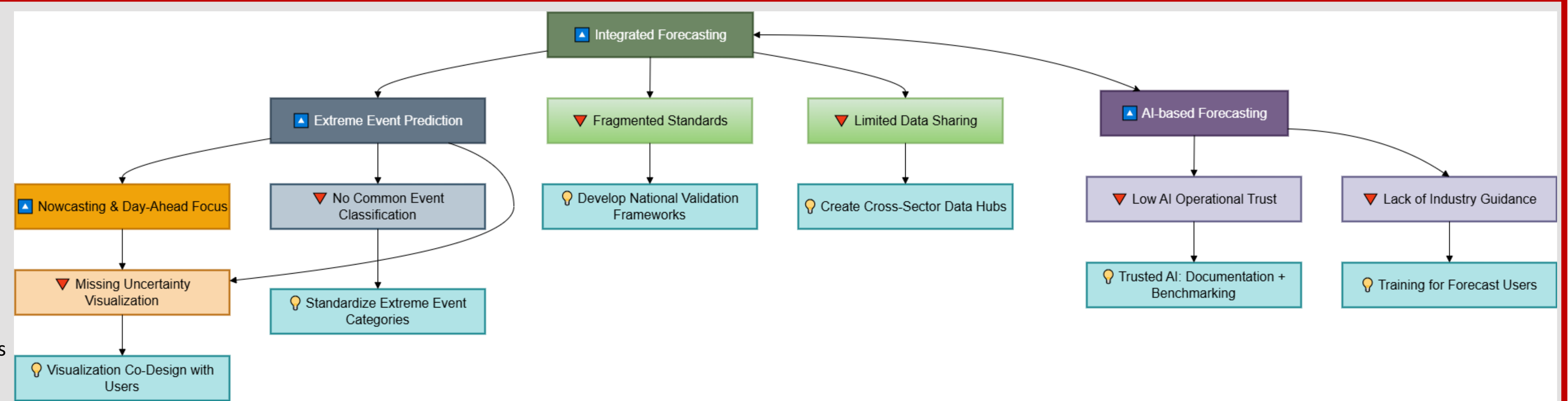


Figure 6. Connected system view of stakeholder priorities, implementation barriers, and proposed solutions. Clear pathways emerge to close the gap between needs and practice.

RECOMMENDATIONS & OUTLOOK

Short-Term Actions (0–12 months)

- Standardize extreme event classification for operational use
- Develop national validation frameworks tailored to complex terrain
- Create uncertainty communication guidelines based on IEA Wind practices
- Document AI systems and establish benchmarking with traditional forecasts

Mid-Term Directions (1–2 years)

- Integrate AI with domain expertise in forecasting products
- Establish cross-border data protocols, especially in Alpine regions
- Launch joint evaluation initiatives (e.g. shared Jupyter-based tools)

Long-Term Outlook (2+ years)

- Operationalize probabilistic forecasts in markets and grid planning
- Enable AI-based real-time decision support for system operators
- Scale integrated forecasting systems across sectors (power, heat, transport)

Next Steps (Workshops 2025)

- Summer: "Extreme Power System Events" – definitions, risks, mitigation → international Task 51 workshop
- Autumn: "Integrated Forecasting" – multi-scale, uncertainty, AI trust

